

TØ 4 – Mikroøkonomi 2 – 2024

Problem 1

Suppose there is a single good that is owned by a single seller who values it at $c > 0$ (if he keeps the good, he can consume it and get a payoff of c).

There is a single buyer who has a small transportation cost $k > 0$ to get to and back from the seller's store, and she values the good at $v > c + k$.

The buyer first decides whether to make the commute or stay at home. In case she stays at home, she does not buy the good and receives a payoff of 0.

If the buyer commutes to the store, the seller can make her a take-it-or-leave-it price offer $p \geq 0$. The buyer can then accept the offer, pay p and get the good, or she can walk out and not buy the good.

Assume that c , v , and k , are common knowledge.

- (a) Draw the game tree of this game.
- (b) What is the best response of the buyer at the node at which she decides whether to accept or reject the seller's offer?
- (c) Find the subgame perfect equilibrium of the game and show that it is unique. Is it Pareto Optimal?
- (d) Are there Nash equilibria that yield a higher payoff to both players as compared to the subgame perfect equilibrium you found in (b) above?

a)

1: The buyer

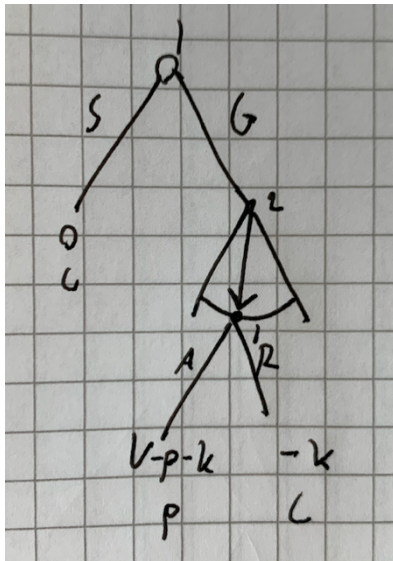
2: The seller

S: Stay home

G: Go to the store

A: Accept

R: Reject



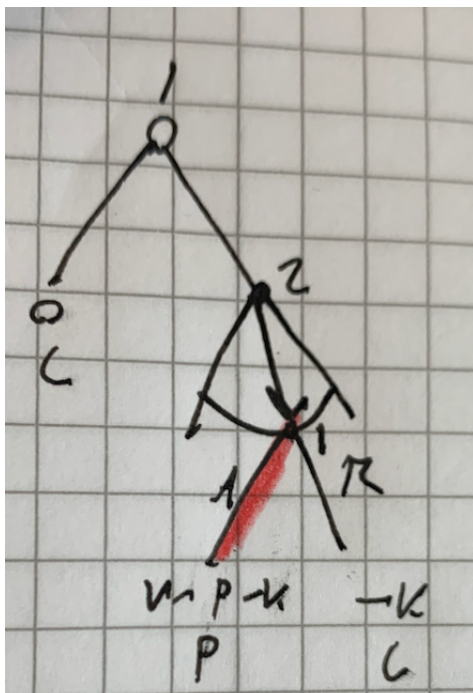
b.

I split this into two scenarios:

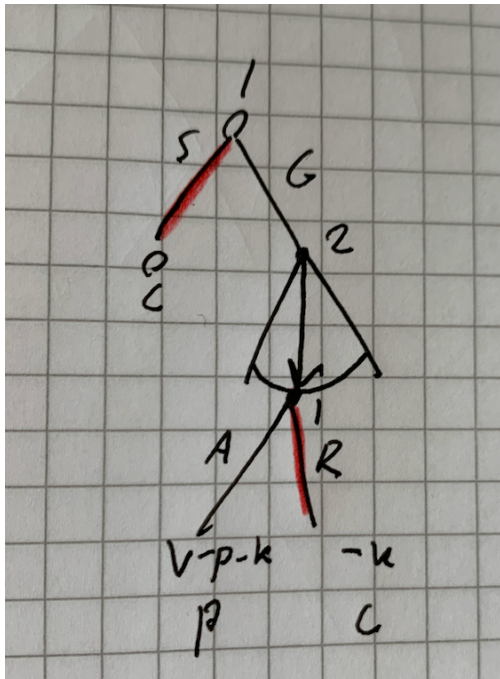
$$v - p - k \geq -k \Rightarrow v \geq p$$

$$v - p - k < -k \Rightarrow v < p$$

When $v \geq p$, the buyer should accept the offer:



When $v < p$, the buyer should reject the seller's offer



c)

From b), we know that the buyer will accept if $v \geq p$

Therefore, seller will choose $p = v$ since this is the highest price the buyer will accept

This implies that the buyer's payoff will be: $v - p - k = v - v - k = -k$

Seeing this, the buyer will decide to stay at home since $-k < 0$ and equilibrium will therefore be

$$(0, c)$$

Equilibrium is then

$$SPNE = \{(S, A)\}$$

It is not Pareto optimal since

Consider the scenario where:

$$c < p < v - k$$

This implies that the payoff in this Nash equilibrium will be

$$(v - p - k, p)$$

Instead of the utility before which was

$$(0, c)$$

It has a higher pay off because

$$(v - p - k, p) > (0, c)$$

This is an example of the fact that the subgame is not Pareto Optimal.

d)

If $p' \in (c, v - k)$ and player 1 (the buyer) will accept any offer: $p' \geq p$, player 2's BR is to offer p' and player 1 will therefore G since $p' < v - k$

Payoffs are

$$(v - p' - k, p') > (0, c)$$

- (e) Now assume that before the game is played, the seller can, at a small cost $\epsilon < (v - c - k)$, send the buyer a postcard that commits the seller to a certain price at which the buyer can buy the good (e.g., "bring this coupon and get the good at a price p "). Would the seller choose to do so? Justify your answer with an equilibrium analysis.

e)

Imagine the seller sends a postcard which means that the price of the good is

$$p' = v - k - \epsilon$$

Buyer will – if she accepts – get a payoff of

$$v - p' - k = v - (v - k - \epsilon) - k = \epsilon > 0$$

Seller will receive payoff (since we know from the information that the buyer values the good: $v > c + k$)

$$p' - \epsilon = v - k - \epsilon - \epsilon = v - k - 2\epsilon > c$$

For small epsilon

Problem 2

It is July 1, siblings Anne (A) and Bo (B) are arguing over how to divide the remainder of their left-over candy from the traditional caramel toss on the 9th graders' last day of school. They decide on the following rules. On July 1, A will make B an offer $(s_A^{(1)}; s_B^{(1)})$ where $s_A^{(1)} + s_B^{(1)} = 1$. If B accepts, then A gets a share $s_A^{(1)}$ of the candy, and Olivia gets $s_B^{(1)}$. If B refuses, however, then no one gets any candy on July 1. In this case, on July 2, B gets to make A an offer $(s_A^{(2)}; s_B^{(2)})$ where $s_A^{(2)} + s_B^{(2)} = 1$. If A accepts, then A gets a share $s_A^{(2)}$ of the candy, and B gets $s_B^{(2)}$. If A refuses, however, then no one gets any candy on July 2. In this case, on July 3, A gets to make B an offer $(s_A^{(3)}; s_B^{(3)})$ where $s_A^{(3)} + s_B^{(3)} = 1$. If B accepts, then A gets a share $s_A^{(3)}$ of the candy, and B gets $s_B^{(3)}$. If B refuses, however, then no one gets any candy on July 3. In this case, on July 4, B gets to make A an offer $(s_A^{(4)}; s_B^{(4)})$ where $s_A^{(4)} + s_B^{(4)} = 1$. If A accepts, then A gets a share $s_A^{(4)}$ of the candy, and B gets $s_B^{(4)}$. If A refuses, however, then all the candy is thrown away by their parents.

There is a catch. Both A and B are impatient for candy so they discount candy tomorrow. For A, each day, one candy tomorrow is worth only δ_A candy today where $0 < \delta_A < 1$. For B, each day, one candy tomorrow is worth only δ_B candy today where $0 < \delta_B < 1$. You may assume that A and B accept an offer whenever he or she is indifferent.

- What do you expect to happen, and what allocation of candy would arise if $\delta_A = \delta_B$? Explain your reasoning carefully.
- Compare your answer when $\delta_A = \delta_B = \frac{3}{4}$, and when $\delta_A = \delta_B = \frac{1}{2}$. Intuitively, why is the impatience of both good for one and bad for the other.
- Now suppose that Anne becomes more impatient so that $\delta_A < \delta_B$. Does this change increase or decrease A's share of the candy relative to what it was when $\delta_A = \delta_B$? Give an intuition.

a)

Since this is a finite horizon bargain with an even number of periods—T is even case—the equilibrium split is $(s_A^{(1)}, s_B^{(1)}) = (1 - \delta + \delta^2 - \delta^3, \delta - \delta^2 + \delta^3)$

Since in the last period, player B can make the offer to keep it all to himself: $(0, \delta^3)$

This implies that in period $T - 1$, player A should offer $\delta^2(1 - s_A) = \delta^3$ in order to make player B indifferent, hence the payoffs in this period is $(\delta^2 s_A, \delta^2(1 - s_A))$ and so on.

b)

When $\delta_A = \delta_B = \frac{3}{4}$:

$$s_A^{(1)} = 1 - \delta + \delta^2 - \delta^3 \approx 0.39$$

$$s_B^{(1)} = \delta - \delta^2 + \delta^3 \approx 0.61$$

And $\delta_A = \delta_B = \frac{1}{2}$

$$s_A^{(1)} \approx 0.62$$

$$s_B^{(1)} \approx 0.38$$

It suggests that when δ is high—high patience, tomorrow's payoff is close to being worth as much as today's—B can exploit the fact that he makes the last move, offering a larger share of the candy to himself in equilibrium. Vice versa, when δ is low, meaning high impatience, player A can exploit the fact that she makes the first move

c)

When $\delta_A < \delta_B$, player A is more impatient than player B. Therefore, A's fraction of the candy will decrease. Intuition is that tomorrow's candy has a smaller value for player A than player B compared to today's value, and player B can thus—compared to before—offer a smaller fraction which player A would accept because of her impatience.

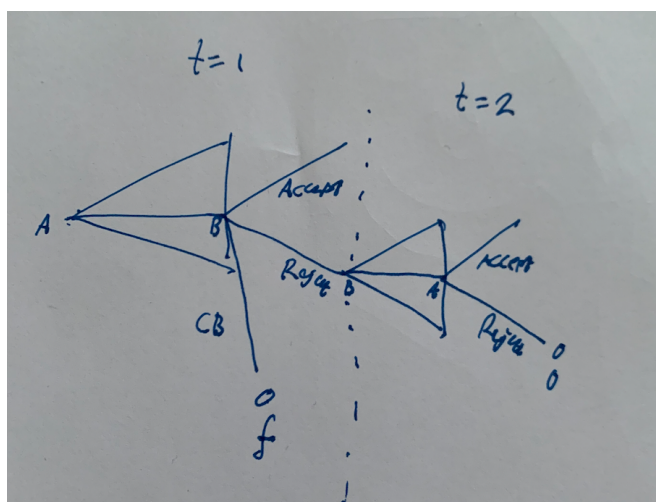
Problem 3

Anne and Bo are still trying to split the stash of candy, normalized to 1, of the previous exercise. This time they play a two-period bargaining game with alternating offers and identical discount factor δ . In particular, Anne offers a split in period 1, while Bo makes his proposal in period 2 (if the game does not end in period 1). The game has the following additional feature: Bo has made an agreement with his mom, who believes that candies are particularly bad sweets, that he can get a chocolate bar if he delivers the entire candy stash. Due to this agreement, Bo can, besides saying Yes or No, choose the chocolate as an outside option. The chocolate bar is worth a fraction f ($0 < f < 1$) of the candy stash to Bo. If Bo chooses the outside option, then Anne gets nothing and the game terminates. The parameter f is commonly known by both players. The game ends when a proposal is accepted, or the outside option is chosen, or at the end of period 2. If Anne and Bo fail to agree and the outside option is not chosen, nobody gets anything as the candy is thrown out by the parents.

- Draw the game tree.
- Find the subgame perfect equilibrium as a function of the parameters f and δ .

a)

Where CB is candy bar



b)

If $f > \delta$

The candy bar option is more attractive for Bo as it gives him a payoff of f which is larger than offer Bo would get in the first period before, δ

Hence, Anna should make an offer which makes Bo indifferent, i.e., the split $(1 - f, f)$

If $f < \delta$

Since the candy bar option is less than δ , it does not matter for the equilibrium: Anna should make the offer $(1 - \delta, \delta)$ as Bo would never accept this and accept the candy bar.